Bridge: Leveraging Vision Foundation Models for Efficient Cross-Domain Remote Sensing Segmentation

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***Abstract*—Adapting large-scale vision foundation models (VFMs), pretrained on natural images (NIs), to remote sensing (RS) downstream tasks such as domain adaptation semantic segmentation (DASS) through parameter-efficient fine-tuning (PEFT) is a promising yet challenging direction. However, due to the large domain gap between NI and RS images (RSIs), as well as the inherent intradomain variations in the RS DASS task, existing PEFT methods struggle to achieve satisfactory adaptation. To address these challenges, we propose Bridge, a framework with small trainable parameters specifically designed for RS DASS. Bridge introduces a set of learnable domain-invariant tokens (DITs) that capture transferable knowledge shared across NI and RSI domains. To complement this, a spatial prior module (SPM) is proposed to extract multiscale spatial priors that provide auxiliary guidance for DIT adaptation (DITA), which produces domain-specific tokens. Finally, domain-aware injection (DAI) injects the domain-specific tokens into intermediate VFM features, aligning them with RS distributions and facilitating effective DASS task adaptation. With significantly fewer trainable parameters, Bridge efficiently transfers NI-pretrained VFMs to the RS DASS task. Extensive experiments demonstrate that Bridge substantially outperforms state-of-the-art PEFT approaches. Remarkably, compared with the current SOTA, Bridge achieves a significant improvement in mIoU. The code will be released at Bridge.**

***Index Terms*—Domain adaptation semantic segmentation (DASS), parameter-efficient fine-tuning (PEFT), remote sensing (RS), vision foundation models (VFMs).**

I. Introduction

REMOTE sensing (RS) techniques are widely applied across diverse visual tasks, including image classification [1], [2], [3], [4], object detection [5], [6], [7], and semantic segmentation [8], [9], [10], [11]. Among these, semantic segmentation aims to assign an accurate class label to each pixel, enabling pixel-level target recognition in RS images (RSIs). Owing to its broad applicability in urban planning, flood control, and environmental monitoring [12], RS semantic segmentation has attracted increasing research interest and remains an active field of study.

In recent years, deep learning-based methods have greatly advanced semantic segmentation, leading to many state-of-the-art architectures, such as fully convolutional neural networks [13], [14], [15] and Transformer-based models [16], [17], [18]. However, their effectiveness strongly depends on the distributional similarity between training and test data. When a domain gap exists between datasets, performance deteriorates significantly [19], [20], [21]. In RS scenarios [22], this domain shift is particularly pronounced due to variations in geographic regions, imaging conditions, and sensor types, resulting in limited generalization of existing methods. To mitigate this issue and establish effective mappings between source and target domains, domain adaptation semantic segmentation (DASS) has emerged as an important research direction.

With the rapid progress of pretraining techniques, large-scale vision foundation models (VFMs) [23], [24], [25], [26], [27], [28], [29], trained on massive natural image (NI) datasets, have achieved remarkable success across a wide range of computer vision tasks. In the RS domain, VFMs [30], [31], [32], [33] can also be leveraged to boost downstream performance by exploiting large volumes of RS data. However, as VFMs grow in scale and capacity, full fine-tuning becomes increasingly impractical: updating and storing an entire copy of model parameters for each downstream task incurs high computational and storage costs.

To address this, parameter-efficient fine-tuning (PEFT) [34], [35], [36], [37] has emerged as a promising paradigm, striking a balance between adaptation capability and efficiency. By updating only a small fraction of parameters, PEFT alleviates the storage and training burden while maintaining competitive performance. Nevertheless, existing PEFT methods often fail when applying VFMs to RS DASS tasks. The main challenge arises from both the substantial interdomain gap between NI and RSIs and the intradomain variations within RS datasets themselves [22], [38], [39], [40], [41]. Unlike NI, where objects usually appear in consistent orientations and contexts, RSI contains objects with diverse scales, complex spatial structures, and arbitrary orientations. Furthermore, differences in geography, imaging conditions, and acquisition equipment exacerbate intradomain variability, making RS DASS significantly more challenging than its NI counterpart. Consequently, VFMs (hereafter referring specifically to those pretrained on

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NI) often underperform or even collapse when directly adapted to RS tasks.

Intuitively, adapting NI-pretrained VFMs to RS DASS is less effective than adapting models pretrained directly on RSI. While pretraining from scratch on RSI could improve performance, this approach is prohibitively expensive, time-consuming, and fails to exploit the rich semantic knowledge encoded in VFMs. This raises a central question: is it truly necessary to pretrain models from scratch for every new domain? Instead, as outlined in Section III, a more efficient strategy is to bridge VFMs to RS domains by incorporating lightweight adaptations that enable effective cross-domain knowledge transfer and facilitate RS DASS.

To this end, we propose Bridge, a lightweight PEFT framework that enhances frozen VFMs without requiring full parameter updates. Bridge introduces three key components: 1) spatial prior module (SPM) to extract multiscale spatial priors; 2) domain-invariant token adaptation (DITA) to combine spatial priors with domain-invariant representations for generating domain-specific tokens; and 3) domain-aware injection (DAI) to seamlessly integrate the adapted tokens into intermediate backbone features. Together, these modules enable VFMs to better align with RS distributions and effectively support DASS tasks.

We conduct extensive experiments on DASS benchmarks to validate the effectiveness of Bridge. Compared with existing methods, Bridge achieves substantial performance gains while maintaining parameter efficiency. Beyond empirical results, we further provide in-depth analyses of the design choices and potential of Bridge.

Our contributions are summarized as follows.

1) We identify the limitations of existing PEFT methods when applied to the RS DASS task and highlight the importance of bridging VFMs to RS domains.
2) We propose Bridge, a lightweight PEFT framework that leverages spatial priors, domain-invariant tokens, and DAI to adapt VFMs effectively to RS tasks.
3) We demonstrate that Bridge significantly outperforms state-of-the-art methods on multiple DASS benchmarks, achieving strong performance improvements with minimal trainable parameters.

II. Related Work

A. DA Semantic Segmentation

With the growing need to deploy models on out-of-domain data, cross-domain segmentation [19], [20], [21] has become increasingly important. DASS aims to adapt a model from a labeled source domain to an unlabeled target domain. In remote sensing (RS), the diversity and complexity of data make cross-domain tasks particularly challenging [22]. Numerous DASS methods have been proposed to enhance the generalization ability of segmentation models, which can be broadly categorized into adversarial learning [39], [40] and self-training [22], [38], [41] approaches. However, these methods rely on task-specific strategies and have not yet leveraged VFMs for the RS DASS task, thereby limiting their performance. Exploring how to effectively integrate VFMs into the DASS task to enhance segmentation accuracy is, therefore, a valuable research direction.

B. Vision Foundation Models

Inspired by the success of Large Language Models (LLMs) [42], VFMs have rapidly advanced, producing a series of powerful models. For example, CLIP [25] employs contrastive learning on large-scale image–text pairs to learn robust visual representations; MAE [26] adopts masked image modeling to capture latent representations; EVA02 [29] integrates masked image modeling pretraining with CLIP's vision features; and DINOv2 [28] achieves strong performance through large-scale unsupervised pretraining. These VFMs exhibit impressive results across diverse downstream tasks. However, adapting VFMs to RS downstream tasks remains challenging due to the substantial domain gap between natural and RSIs, as well as task-specific gaps (e.g., in DASS). Consequently, many works have begun to develop geospatial foundation models (GFMs) [30], [31], [32], [33] tailored for RS. For instance, RemoteCLIP [43] extends contrastive pretraining to RS data, SatMAE [44] adapts the MAE framework to RS imagery, and RingMo [45] proposes a masked self-supervised learning strategy for RS. While these approaches highlight the potential of GFMs, retraining such large-scale models from scratch in RS is prohibitively expensive. Thus, efficiently bridging NI-pretrained VFMs to RS downstream tasks is an important and underexplored problem.

C. Parameter-Efficient Fine-Tuning

In natural language processing (NLP), PEFT methods such as adapters [34], [46] were introduced to adapt large pretrained models to downstream tasks with only a small number of trainable parameters. Following their success in NLP, PEFT approaches have been extended to vision tasks and have attracted considerable attention [47], [48]. For example, vision prompt tuning (VPT) [35] introduces learnable prompt tokens into the input sequence, while AdaptFormer [49] replaces the MLP blocks in Transformer encoders with lightweight adaptation layers. Currently, cross-domain works also focus on PEFT, such as [50], which is the first prompt tuning method for domain adaptation (DA) and [51] is the first work leveraging PEFT to fine-tune VFMs for domain generalization semantic segmentation (DGSS). After that, [52] proposed a concept-guided domain generalization approach for DGSS. [53] proposed a domain-generalized token linking approach to mine domain-invariant information from VFMs for improving the performance in unseen target domains. In the RS field, the exploration of PEFT is also vigorous. UPetu [36] addresses storage for dense prediction tasks. Tea [37] uses an adapter network and top-down guidance for detection and segmentation tasks. CrossEarth [54] and Earth-Adapter [55] leverage PEFT to fine-tune VFMs for DGSS. However, CrossEarth and Earth-Adapter primarily focus on DGSS, i.e., improving generalization toward unseen target domains without target exposure, which differs from the setting of DASS that assumes access to target data during adaptation. In that case, PEFT

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methods remain largely unexplored in the context of fine-tuning VFMs to the RS DASS task. Existing PEFT methods fail to adequately bridge the substantial domain gap from NI to RSI, limiting the transferability of VFMs.

In summary, prior works in RS DASS have primarily relied on specific strategies (adversarial or self-training), while research on VFMs has demonstrated their potential, but existing PEFT methods often fail to apply VFMs to RS DASS tasks, due to the interdomain gap between NI and RSI and the intradomain gap between RS DASS source and target domains. This motivates our work: we propose Bridge, a PEFT-based framework that introduces domain-invariant tokens and spatial priors, progressively injecting them into frozen VFMs to effectively align NI knowledge with RS DASS tasks.

III. Problem Setup

Transferring large-scale pretrained VFMs from NIs to RS downstream tasks such as DASS [22] is a promising yet highly challenging problem.

Formally, let $D = \{d_1, d_2, d_3, \ldots\}$ denote a collection of domains, where each domain d contains images $x \in \mathbb{R}^{C_d \times H_d \times W_d} \sim p_d(x)$. Denote NIs as the source domain d_s and RSIs as the target domain d_t. In practice, both d_s and d_t often contain inner-domain gaps (e.g., day versus night conditions in d_s, different satellite sensors, or regional differences in d_t). Thus, in d_t, we typically require downstream adaptation tasks DA to transfer knowledge from an inner source domain t_s to an inner target domain t_t.

Given a pretrained VFM with weights ω_s obtained on d_s, our objective is to adapt it for downstream tasks on d_t, namely:

DASS: Transferring from t_s to unlabeled t_t, using both labeled data $\mathcal{Y}_{t_s}$ and unlabeled data $\mathcal{X}_{t_t} = \{x_{t_t}^i\}_{i=1}^{N_{t_t}}$.

The goal is to efficiently learn task-specific weights ω_t^{da} by leveraging knowledge encoded in ω_s, while bridging both the large interdomain gap (between NI and RSIs) and the intradomain gaps within the RS target domain.

Conventional approaches [33], [43], [44], [45] assume oracle initialization with weights ω_t pretrained directly on the target domain, then fine-tuned for a specific downstream task

$$\omega_t^{(task)} \approx \omega_t + \Delta_{task}. \tag{1}$$

However, this strategy requires expensive pretraining for every new target domain d_t to obtain ω_t and full fine-tuning of the VFM backbone, which is inefficient.

PEFT methods (e.g., LoRA [34], Adapters [46]) alleviate this problem by introducing lightweight trainable modules

$$\omega_t^{(task)} \approx \omega_t + \Delta_{\text{PEFT}}^{task} \tag{2}$$

where only a small fraction of parameters are updated. In practice [22], [39], [40], instead of pretraining on d_t, we can reuse ω_s from d_s and apply PEFT for downstream adaptation. Yet, under large domain shifts (e.g., NI $\rightarrow$ RS), directly using current PEFT methods often fails, suffering from representation collapse.

To address this, we hypothesize that effective transfer under both interdomain and intradomain gaps requires a two-level bridging mechanism

$$\omega_t^{(task)} \approx \omega_t + \Delta_{bridge} + \Delta_{task} \tag{3}$$

where Δ_{bridge} specifically accounts for domain discrepancies (NI $\rightarrow$ RS) and Δ_{task} further accounts for intradomain variations in DASS task, both are realized via lightweight PEFT modules.

IV. Method

A. Overview

Directly applying NI-pretrained VFMs to RS tasks results in poor adaptation due to substantial domain discrepancies. To address this, we propose Bridge, a lightweight PEFT framework that enhances frozen VFMs by introducing spatial priors and domain-invariant tokens, which are injected into backbone features.

As illustrated in Fig. 1, the input image follows two parallel processing pathways. On the one hand, it undergoes standard preprocessing operations, such as patch embedding and positional encoding, to produce the initial input sequence for the VFM. On the other hand, the image is processed by the SPM to generate spatial prior features f_{sp}. At each layer of the VFM, the output features from the previous layer, f_i, are refined in two steps. First, the DITA module leverages the spatial prior features f_{sp} to adapt the domain-invariant tokens to produce domain-specific tokens. Then, the DAI module injects these domain-specific tokens into the f_i, yielding refined representations $\bar{f}_i$ that serve as inputs to the next layer. Finally, the refined features from the last layer are used as the VFM output, which is passed to the decoder head to generate pixel-level predictions.

B. Bridge

Formally, given an input image x, the feature propagation is defined as

$$\begin{aligned} f_1 &= L_1(\text{Embed}(x)) \\ f_{\text{out}} &= f_i + \text{Bridge}(f_{i+1}, x) \\ f_{\text{out}} &= f_N + \text{Bridge}(f_N, x) \end{aligned} \tag{4}$$

where all backbone layers L_i are frozen, and only the parameters within the Bridge are learnable. Bridge is composed of three modules.

1) Spatial Prior Module: RSIs exhibit strong spatial regularities (e.g., man-made structures, parcel layouts) that vanilla transformers poorly capture. Inspired by prior work showing that convolutions provide effective inductive biases for spatial modeling, SPM extracts multiscale spatial priors to serve as auxiliary knowledge for token adaptation.

Specifically, the SPM begins with a standard convolutional stem, which consists of three stacked convolutional layers followed by a max-pooling layer. All convolutional layers use a kernel size of 3×3, padding of 1, and a stride of 1, except for the first convolution, which has a stride of 2. Following the stem, the module contains three sequential convolutional blocks, each with a 3×3 kernel, stride 2, and padding 1. These blocks progressively reduce the spatial resolution while doubling the number of channels, producing feature maps at three different scales. Finally, each multiscale feature map

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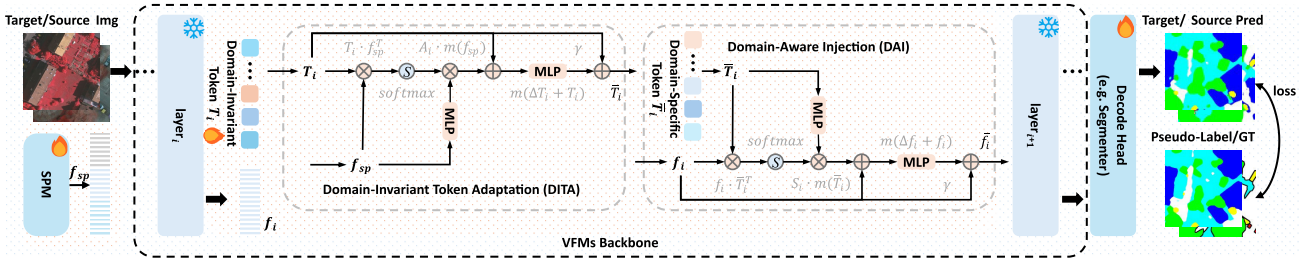


Fig. 1. Overview of proposed Bridge. Bridge is composed of three main components: the SPM, the DITA, and the DAI. The SPM extracts spatial prior features f_{sp} from the input image. For each backbone layer, the DITA leverages both the spatial prior features f_{sp} and learnable domain-invariant tokens $T_i \in \mathbb{R}^{m\times d}$ to generate domain-specific tokens $\bar{T}_i$. Subsequently, the DAI integrates the domain-specific tokens $\bar{T}_i$ with the corresponding feature f_i to produce enhanced features $\bar{f}_i$. To improve efficiency, all MLP parameters are shared across layers, thereby reducing the total number of trainable parameters.

is projected via a 1×1 convolution to match the channel dimension d of the VFM

$$f_{sp}^1, f_{sp}^2, f_{sp}^3 = \mathrm{SPM}(x), \quad f_{sp}^i \in \mathbb{R}^{\frac{H\times W}{2^{i+2}}\times d}. \tag{5}$$

After that, each feature map is flattened, positionally encoded, and augmented with level embeddings

$$\hat{f}_{sp}^i = \mathrm{flatten}\left(f_{sp}^i\right) + \mathrm{pos_embed}_i + \mathrm{level_embed}_i \tag{6}$$

which are concatenated into a spatial prior token sequence

$$f_{sp} = \mathrm{concat}\left(\hat{f}_{sp}^1, \hat{f}_{sp}^2, \hat{f}_{sp}^3\right). \tag{7}$$

These priors provide domain-specific structural cues that later interact with invariant tokens.

2) Domain-Invariant Token Adaptation: Although VFMs pretrained on NI capture rich semantics, their representations are biased toward natural-image statistics and thus misaligned with RSIry. To mitigate this, we introduce a set of learnable domain-invariant tokens $T_i \in \mathbb{R}^{m\times d}$ for each layer i. These tokens encode transferable knowledge shared across NI and RS domains (see Section VI-C for detailed discussion).

To adapt them for RS tasks, DITA enables tokens to absorb spatial priors from SPM. First, we compute a similarity map

$$A_i = \mathrm{Softmax}\left(\frac{T_i f_{sp}^T}{\sqrt{c}}\right) \tag{8}$$

and update tokens as

$$\begin{aligned}\Delta T_i &= A_i \cdot \mathrm{MLP}_1\left(f_{sp}\right),\\ \bar{T}_i &= T_i + \gamma \cdot \mathrm{MLP}_2\left(T_i + \Delta T_i\right).\end{aligned} \tag{9}$$

Here, $\bar{T}_i$ inherits domain-invariant knowledge while integrating RS-specific priors, serving as a bridge between NI and RS feature spaces.

3) Domain-Aware Injection: Finally, DAI injects adapted tokens into intermediate VFM features, progressively aligning them with RS distributions. At layer i, the feature f_i interacts with $\bar{T}_i$ through feature–token similarity

$$S_i = \mathrm{Softmax}\left(\frac{f_i \bar{T}_i^T}{\sqrt{c}}\right), \quad S_i \in \mathbb{R}^{n\times m}. \tag{10}$$

Refined features are computed as

$$\Delta f_i = S_i[:, 2:m] \cdot \mathrm{MLP}_3\left(T_i[:, 2:m]\right) \tag{11}$$

$$Bridge\left(f_i, x\right) = f_i + \gamma \cdot \mathrm{MLP}_4\left(f_i + \Delta f_i\right). \tag{12}$$

This DAI ensures that features are progressively adapted to RS distributions while retaining semantic knowledge from NI pretraining. To reduce overhead, the MLPs in DITA and DAI are shared across layers.

4) Optimization Objective: The final model consists of a frozen VFM encoder $\mathcal{E}_{\omega_s}$, a task-specific head h_θ with parameters θ, and the learnable Bridge parameters $\Delta_{bridge}, \Delta_{task}$. Training minimizes the downstream task loss

$$\min_{\Delta_{bridge},\Delta_{task},\theta} \mathcal{L}_{task}\left(h_\theta\left(\mathcal{E}_{\omega_s}\left(x; \Delta_{bridge}, \Delta_{task}\right)\right), y\right). \tag{13}$$

For DASS, training uses labeled source data and additionally incorporates unlabeled target samples

$$\left\{\mathcal{Y}_{t_s}, \mathcal{X}_{t_t}\right\} = \left\{\left(x_{t_s}^i, y_{t_s}^i\right), x_{t_t}^j\right\}_{i=1,j=1}^{N_{t_s},N_{t_t}}. \tag{14}$$

In this work, the proposed framework adopts a standard self-training paradigm [20], [22] for domain-adaptive semantic segmentation, where an EMA teacher produces pseudo labels for target-domain images and the student model is jointly supervised on both labeled source data and pseudo-labeled target data.

Specifically, given an input image x, the segmentation head produces pixel-wise class probabilities $\hat{y} \in \mathbb{R}^{H\times W\times C}$

$$\hat{y} = h_\theta\left(\mathcal{E}_{\omega_s}\left(x; \Delta_{bridge}, \Delta_{task}\right)\right). \tag{15}$$

The segmentation loss is defined as a pixel-wise cross-entropy loss

$$\mathcal{L}_{seg} = -\frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W}\sum_{c=1}^{C} y_{hw}^{(c)} \log \hat{y}_{hw}^{(c)} \tag{16}$$

where $y \in \{0,1\}^{H\times W\times C}$ denotes either a one-hot source-domain label or a hard pseudo label generated by the EMA teacher, and C is the number of categories.

The overall DASS loss becomes

$$\mathcal{L}_{dass} = \sum_{\mathcal{Y}_{t_s}} \mathcal{L}_{seg}\left(x_{t_s}^i, y_{t_s}^i\right) + \sum_{\mathcal{X}_{t_t}\cup p_t} \mathcal{L}_{seg}\left(x_{t_t}^j, p_t^j\right) \tag{17}$$

where $p_t = \{p_t^j\}_{j=1}^{N_{t_t}}$ denotes the pseudo labels predicted by the EMA teacher.

V. Experiment

A. Experimental Setup

1) Datasets and Benchmark: To evaluate the proposed method's performance in cross-domain RSI segmentation

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TABLE I
DATASET OVERVIEW.* INDICATES THAT IMAGE SIZES VARY WITHIN THE DATASET, HENCE THE VALUES PROVIDED REPRESENT AVERAGES. † DENOTES THE NUMBER OF IMAGES AFTER CROPPING

Dataset	Resolution	Classes	Crop / Stride	Train split	Valid split
ISPRS Benchmarks					
Potsdam	6000×6000	6	512 / 512	24 (2904†)	14 (1694†)
Vaihingen	2906×2273*	6	512 / 256	16 (1296†)	17 (440†)
LoveDA Benchmarks					
Urban	1024×1024	7	-	1156	667
Rural	1024×1024	7	-	1366	992

tasks, we selected three benchmark datasets: Potsdam [56], Vaihingen [56], and LoveDA [57]. Table I presents the image resolution, number of categories, and Train/Valid splits for different datasets. It also shows the cropping size and stride used during data preprocessing.

The Potsdam and Vaihingen datasets are part of the ISPRS 2D semantic segmentation benchmarks [56]. The Potsdam (POT) dataset comprises 38 RSIs with a resolution of 6000 × 6000 pixels and a ground sampling distance of 5 m. It features three different image modalities: IRRG, RGB, and RGBIR, where the first two modalities have three channels, and the last one has four. In this study, we primarily utilize the first two modalities. The Vaihingen (VAI) dataset contains 33 RSIs with resolutions ranging from 1996 × 1996 to 3816 × 2550 pixels, with a ground sampling distance of 9 cm. This dataset includes only one image modality: IRRG. Both datasets share six common classes: impervious surfaces, buildings, low vegetation, trees, cars, and clutter/background.

To reduce computational overhead, we cropped the images to a smaller size of 512 × 512 pixels. For the POT and VAI datasets, we used cropping strides of 512 and 256, resulting in 4598 and 1696 images, respectively. Subsequently, we split the POT and VAI datasets into training and testing sets, following previous works [22]. In the POT dataset, the number of images in the training and testing sets is 2904 and 1694, respectively, while in the VAI dataset, these numbers are 1296 and 440. We established four cross-domain RS semantic segmentation tasks.

1) Potsdam to Vaihingen (P2V).
2) Vaihingen to Potsdam (V2P).

The LoveDA [57] dataset was recently proposed to address semantic segmentation and DA challenges in RS. It consists of 5987 high-resolution (1024 × 1024) RSIs sourced from Nanjing, Guangzhou, and Wuhan. The LoveDA dataset contains two distinct domains: urban and rural, aimed at challenging the model's generalization capability between different geographical elements.

Within the dataset, there are 1883 urban images, which are further divided into 1156 training samples and 677 validation samples as mentioned in [57]. In the rural domain, there are 2358 images, with 1366 for training and 992 for validation. For the LoveDA dataset, we designed a cross-domain RS semantic segmentation task.

1) Rural → Urban (R2U).

2) Evaluation Metrics: In this study, we adopt the intersection over union (IoU) as an evaluation metric, following previous methods in the field of RS semantic segmentation DA (UDA). The specific calculation formulas are as follows:

$$\text{IoU} = \frac{TP}{TP + FP + FN}. \tag{18}$$

In equations [see (18)], TP represents true positives, FP denotes false positives, and FN indicates false negatives. The IoU, also known as the Jaccard index, effectively reflects the accuracy and reliability of model performance in semantic segmentation tasks.

3) Implementation Details: We use MMSegmentation[1] [58] as our training and evaluation framework. To improve computational efficiency, automatic mixed precision (AMP) with BF16 format is consistently applied. We employ Segmentor [59] as our decoder, which is a highly efficient and effective algorithm for semantic segmentation tasks. For backbone, we utilize the DINOv2 [28] series models with patch size 16 to extract features, which are subsequently utilized as input for the Segmentor. During training, we utilize AdamW as the optimizer, with a learning rate of $1e^{-4}$ for the decoder, and $1e^{-4}$ for the relevant parameters in Bridge, along with a poly learning rate schedule and a linear warmup of 1500 steps. Training is conducted for 40k steps with a batch size of 2 on a single NVIDIA A100 GPU. Input images were standardized to a resolution of 512 × 512 pixels and subjected to a comprehensive augmentation pipeline, including random resizing, cropping, flipping, and photometric distortion. Within the DACS [20] training framework, we set α to 0.968. For Bridge, we set $\gamma = 0.15$. Thanks to our streamlined training configuration and reduced number of trainable parameters, Bridge can fine-tune VFMs like DINOv2-Large efficiently on a single NVIDIA A100 GPU within ~4 h for superior cross-DA ability.

B. Cross-Domain Performance Comparison

1) Comparing Bridge With Existing DASS and PEFT: As shown in Table II, this part conduct a comprehensive comparison between mainstream PEFT methods (Adapter [46], LoRA [34], VPT [35], AdaptFormer [49], Rein [51], CrossEarth [54], Earth-Adapter [55]), existing DASS methods (DAFormer [19] and SiamSeg [22]), and proposed Bridge across three cross-DA tasks: P2V, V2P, and R2U. For fair comparison, all PEFT methods use the DACS [20] training framework. The implementation of all PEFT methods in the table References the implementation of Earth-Adapter [55]. Compared with DASS-specific methods, several existing PEFT approaches exhibit severe performance collapse when applied to RS cross-domain tasks. For instance, LoRA achieves only 17.8% mIoU on P2V and 24.5% mIoU on R2U, which is even lower than the accuracy obtained by simply freezing the DINOv2-L [28] backbone. Similar degradation is also observed with full Fine-Tuning and AdaptFormer. We attribute this phenomenon to the substantial domain gap between NIs and RSIs, which hinders the effective transfer of pretrained

[1]https://github.com/open-mmlab/mmsegmentation

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TABLE II

PERFORMANCE COMPARISON OF THE PROPOSED BRIDGE AGAINST OTHER DASS AND PEFT METHODS UNDER THE P2V, V2P, AND R2U DA SETTING. "+" MEANS THE BACKBONE IS FROZEN. THE BOLD FONT MEANS THE BEST PERFORMANCE

BackBone	Method	Cross-Domain Task			Avg.
		P2V	V2P	R2U	
Previous SOTA UDA methods					
MiT-B5 [17]	DAFormer [19]	51.3	54.6	48.3	52.9
	SiamSeg [22]	55.1	55.7	51.7	54.1
Previous PEFT Methods					
DINOv2-L [28]	Frozen	21.0	7.2	21.5	16.5
	Full Fine-Tune	11.3	16.5	23.1	16.9
	+Adapter [46]	66.4	59.3	55.9	60.5
	+LoRA [34]	17.8	18.8	24.5	20.3
	+VPT [35]	66.2	59.3	55.3	58.9
	+AdaptFormer [49]	12.9	15.3	25.1	18.1
Cross-Domain PEFT Methods					
DINOv2-L [28]	+Rein [51]	60.2	60.9	52.8	57.9
	+CrossEarth [54]	66.8	61.4	53.7	60.6
	+Earth-Adapter [55]	67.7	62.2	55.9	61.9
	+Bridge	**75.2**	**67.7**	**56.5**	**66.4**

knowledge and prevents successful adaptation to RS DA tasks. Nevertheless, four PEFT methods: Adapter, VPT, CrossEarth, and Earth-Adapter, demonstrate relatively robust performance on RS cross-domain tasks. Some PEFT methods, such as LoRA, which injects rank-r updates directly into the Q/K/V projections of VFMs, struggle to adapt to the characteristics of RSIs, often leading to accelerated feature drift, causing catastrophic forgetting of the pretrained features and disrupting the representations of VFMs during fine-tuning, ultimately leading to poor performance. In comparison, the proposed Bridge exhibits remarkable adaptability, effectively transferring knowledge from natural-image pretraining to RS domains and achieving superior cross-domain generalization. Specifically, Bridge attains 75.2% mIoU on P2V, 67.7% mIoU on V2P, and 56.5% mIoU on R2U. Notably, Bridge consistently surpasses the second-best method, Earth-Adapter, by +7.4% on P2V, +5.5% on V2P, and +0.4% on R2U.

2) Comparing Bridge on Per-Class Performance: As shown in Table III, we provide a detailed per-class comparison of Frozen DINOv2-L, SiamSeg, Earth-Adapter, and the proposed Bridge across two benchmarks: P2V and Rural to Urban (R2U).

On the P2V task, compared with Frozen DINOv2-L, the DASS-oriented SiamSeg achieves consistent improvements across all categories. Relative to SiamSeg, both Earth-Adapter and Bridge exhibit substantial gains in the vegetation class, with improvements exceeding 20% mIoU. Notably, Bridge achieves exceptional accuracy in the clutter category, surpassing frozen DINOv2-L by over 70% mIoU, SiamSeg by over 60%, and Earth-Adapter by over 40%. Since clutter is characterized by high intraclass variability and low occurrence frequency, it poses a significant challenge for existing approaches. The superior performance of Bridge in this class highlights its ability to better transfer pretrained knowledge into the RS domain and adapt effectively to the DASS setting.

On the R2U task, a similar trend is observed: Frozen DINOv2-L suffers from performance collapse under large domain shifts. Both Earth-Adapter and Bridge achieve strong results in the Agriculture and Forest classes. However, Earth-Adapter underperforms on categories such as barren and road, likely due to limited exploitation of pretrained representations. In contrast, Bridge maintains strong recognition capability on these challenging categories, further demonstrating its robustness in transferring pretrained knowledge.

Overall, Bridge consistently achieves state-of-the-art performance across most categories, yielding the highest mean IoU (mIoU) on both P2V and R2U tasks.

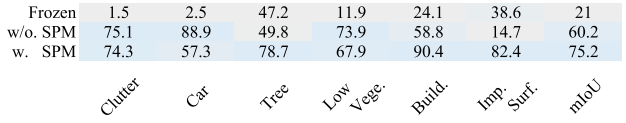


Fig. 2. Effectiveness Ablation of the SPM. Here presents the class-wise IoU of different methods on the cross-domain task P2V. To better illustrate the gains contributed by each component, we employ varying shades of blue and gray to demonstrate the relative performance of the Freeze and Bridge methods. All methods are based on the frozen DINOv2-L.

C. Ablation Study and Analysis

1) Analysis of the Key Components: Effectiveness Ablation of the SPM: Fig. 2 presents the ablation results for the SPM. Note that when SPM is not used, domain-invariant tokens are directly employed in place of domain-specific tokens for DAI refinement. The analysis shows that removing SPM leads to a substantial performance degradation, as observed in Bridge without SPM. This degradation can be attributed to the significant variations in object sizes in RSI, in contrast to the relatively uniform object scales in NIs, which makes Bridge without SPM less effective in recognizing objects across different scales. By incorporating SPM, the model gains access to richer multiscale information, which greatly facilitates the transfer of pretrained knowledge from NI to RSI. Consequently, the model achieves a notable performance improvement. In particular, the Tree and Building categories demonstrate IoU gains of 28.9% and 32.6%, respectively—both of which are classes characterized by large intraclass scale variations in RSI. These results further validate the effectiveness of SPM.

Importance of SPM: While SPM is trained from scratch and early-stage priors may be inaccurate, this has only a minor impact on adaptation. This is because DITA applies residual token refinement rather than overriding the domain-invariant tokens, and DAI employs a similar refinement mechanism, so initial inaccuracies only produce weak modulation and do not destabilize training. The ablation results in Fig. 2 clearly demonstrate that optimizing SPM is critical. This is consistent with the observation that domain gaps in RS largely arise from spatial layout and scale variations. The multiscale spatial priors provided by SPM guide the tokens to capture domain-specific structures, a capability that cannot be achieved by directly using domain-invariant tokens for DAI refinement of VFM features.

Replace SPM with Shallow Features: The SPM serves as an auxiliary model to extract RS-specific spatial features, which interact with domain-invariant tokens in DITA to produce domain-specific tokens. These tokens are subsequently

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TABLE III

PERFORMANCE COMPARISON OF PER-CLASS BETWEEN FROZEN DINOV2-L, SIAMSEG, EARTH-ADAPTER AND BRIDGE ON P2V AND R2U. † DENOTE THE METHOD IS BASED ON THE FROZEN DINOV2-L. THE BOLD FONT MEANS THE BEST PERFORMANCE

Method	Task	Clutter	Car	Tree	Vege.	Build.	Imp.	mIoU	Task	Agri.	Forest	Barren	Water	Road	Build.	Back.	mIoU
Frozen†	P2V	1.5	2.5	47.2	11.9	24.1	38.6	21.0	R2U	22.1	28.7	13.7	21.9	15.6	23.7	22.4	21.5
SiamSeg [22]		13.2	51.1	71.1	40.1	81.6	73.2	55.1		49.1	47.3	47.4	71.3	56.6	52.5	37.6	51.7
Earth-Adapter† [55]		33.6	**61.9**	74.4	66.3	89.5	80.3	67.7		**61.8**	52.8	42.2	**74.6**	54.6	**62.5**	**61.8**	55.9
Bridge†		**74.3**	57.3	**78.7**	**67.9**	**90.4**	**82.4**	**75.2**		59.5	**54.2**	**50.2**	73.6	**57.0**	59.7	41.7	**56.5**

TABLE IV

STUDY ON SPM AND SHALLOW FEATURES. NOTE THAT WE REPLACE THE f_{sp} WITH THE FEATURE f_1 AND THAT THE BACKBONE LAYERS L_1 ARE USED. ALL EXPERIMENTS ARE BASED ON CROSS-DOMAIN TASK P2V

Backbone	SPM	Shallow Features	mIoU(%)
DINOv2 [28] (Large)	-	✓	59.4
	✓	-	75.2

	Clutter	Car	Tree	Low Vege.	Build.	Imp. Surf.	mIoU
Frozen	1.5	2.5	47.2	11.9	24.1	38.6	21
+share (all)	46.4	36.9	64.9	28.4	78.1	60.1	52.5
+share (12)	35.4	41.9	56.6	41.1	79.9	70.9	54.2
+share (6)	62.8	56.1	78.8	67.2	91.4	80.8	72.8
+share (1)	74.3	57.3	78.7	67.9	90.4	82.4	75.2

Fig. 3. Effectiveness Ablation of the Token Sharing on cross-domain task P2V. All methods are based on the frozen DINOv2-L.

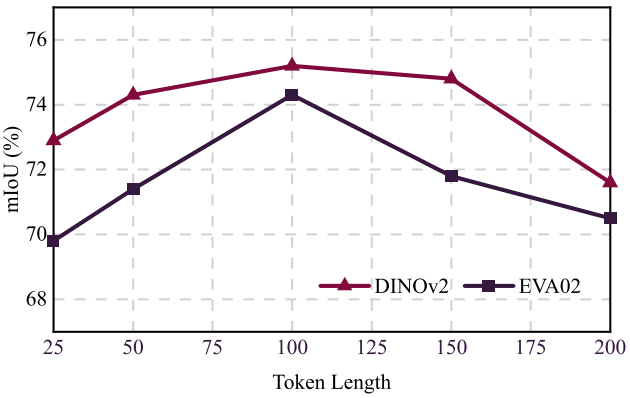


Fig. 4. Study on Domain-Invariant Token length. All methods are based on cross-domain task P2V.

TABLE V

ABLATION STUDIES OF BRIDGE ON DIFFERENT FOUNDATION BACKBONES ACROSS TWO BENCHMARKS: P2V AND V2P. FULL REPRESENTS FULL FINE-TUNE

Backbone	Scale	Method	Params	P2V	V2P	Avg.
CLIP [25]	Large	Frozen	0M	23.8	11.5	14.2
		Full	304.2M	14.5	17.1	13.9
		+Bridge	**8.3**M	**55.3**	**54.2**	**51.2**
EVA02 [29]	Large	Frozen	0M	20.1	6.9	14.0
		Full	304.2M	11.3	16.5	13.9
		+Bridge	**8.3** M	**74.3**	**66.1**	**70.2**
DINOv2 [28]	Large	Frozen	0M	21.0	7.2	14.1
		Full	304.2M	11.3	16.5	13.9
		+Bridge	8.3 M	**75.2**	**67.7**	**71.4**
	Base	+Bridge	4.8M	62.3	59.7	61.0
	Small	+Bridge	**4.5**M	54.7	53.4	58.1

injected via DAI to refine the intermediate features of the VFM. Although SPM consists of simple convolutional layers, it provides crucial RS-specific spatial priors that are not present in the shallow features of the ViT backbone. The ViT shallow features are generated from NI-pretrained patch embeddings, which, due to the large domain gap between NI and RSI, cannot adequately capture RS-specific structural information. In contrast, SPM explicitly learns spatial patterns from RS data, enabling DITA to adapt domain-invariant tokens into domain-specific representations, which are then effectively utilized by DAI to refine VFM features. To validate this, we conducted an experiment replacing f_{sp} with the ViT shallow features f_1, where the backbone layers L_1 are used, in DITA. Table IV shows a notable performance drop compared to using SPM, confirming that shallow features alone are insufficient for capturing the RS-specific spatial priors necessary for effective DA. This demonstrates that even a simple convolutional SPM plays a critical role in bridging the NI-RS domain gap and achieving strong cross-DA performance.

Effectiveness Ablation of the Domain-Invariant Token: To evaluate the effectiveness of the domain-invariant token (DIT) across different layers, we designed a series of controlled experiments in which the DIT was shared between adjacent layers. The results are presented in Fig. 3. The analysis reveals that when all layers share the same DIT, the model achieves a noticeable improvement in per-class performance compared to the frozen baseline. However, when the number of shared layers is reduced to 12, corresponding to two distinct DIT groups, the performance gain becomes marginal. In contrast, further reducing the shared layers to 6, thereby allowing more layers to employ distinct tokens, leads to a substantial performance boost. This improvement is particularly evident in the clutter, tree, and low vegetation categories, suggesting that utilizing different tokens at different layers better captures hierarchical domain-invariant features and contributes to more robust performance.

Study on Domain-Invariant Token length: In this section, we investigate the effect of different lengths of the DIT, denoted as $T \in \mathbb{R}^{m \times c}$, where m ranges from 25 to 200. The results are presented in Fig. 4. As shown, when $m = 25$, the model performance is suboptimal. With increasing m, performance improves and reaches its peak at $m = 100$. However, further enlarging the token length beyond 100 does not yield additional gains; instead, starting from $m = 150$, performance begins to decline. These findings suggest that choosing an appropriate token length is crucial, as excessively short or excessively long tokens hinder the model's ability to effectively capture domain-invariant features.

2) Performance on Different Backbones: We further evaluated Bridge on various backbones, including EVA02, CLIP, and multiple scales of DINOv2 (i.e., Small, Base, and

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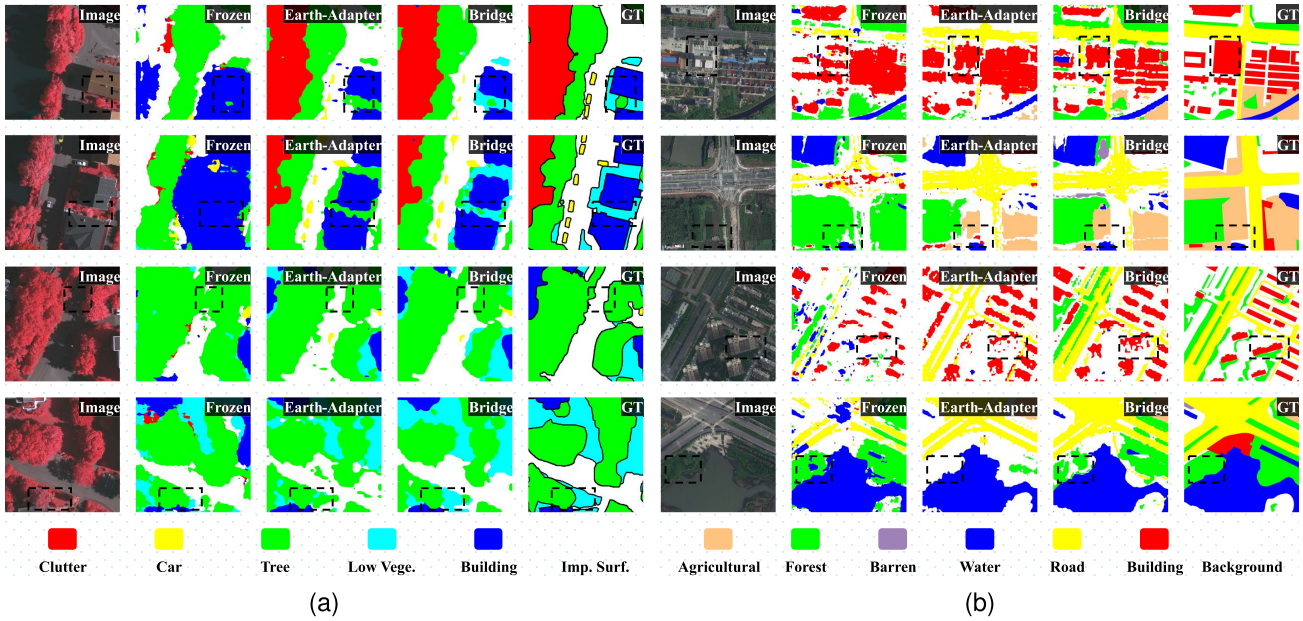


Fig. 5. Visualization of results on cross-domain tasks. (a) P2V. (b) R2U. Note that all methods are based on DINOv2.

Large). The results, summarized in Table V, demonstrate that Bridge consistently improves performance across different architectures and parameter scales. This indicates that Bridge effectively transfers pretrained knowledge without relying on a specific backbone. Notably, when models are pretrained using the same strategy, the performance gains achieved by Bridge increase proportionally with the model size, highlighting its scalability.

3) Prediction Visualization Analysis: Fig. 5 presents the visualization results of Frozen DINOv2-L, Earth-Adapter, and Bridge. As shown, the frozen approach struggles to distinguish different categories, likely due to the substantial domain gap between natural and RSIs, which hinders effective knowledge transfer to downstream RS tasks. With the introduction of PEFT, Earth-Adapter achieves moderate improvements in recognition. In contrast, the proposed Bridge demonstrates highly accurate object delineation, indicating its superior capability in transferring pretrained knowledge and effectively adapting to cross-domain RS tasks.

4) SPM Visualization Analysis: To better understand the behavior of SPM, we visualize its multiscale output features. For each level, we apply channel-wise average pooling to generate spatial heatmaps and overlay them on input images. As shown in Fig. 6, SPM consistently highlights semantically meaningful RS structures (e.g., buildings, road layouts, and vegetation boundaries), demonstrating that it extracts RS-specific spatial priors rather than shallow texture-level cues. These priors provide valuable domain cues for the subsequent token adaptation stage.

5) T-SEN Visualization: To better validate that the feature distributions of the source, target, and inner domains are properly aligned, we visualize the final-layer features from the backbone using T-SNE embeddings [60], as shown in Fig. 7. In the visualization, features from different domains are denoted using dots and stars, respectively. As highlighted by the green circles, features extracted by Bridge form tight and

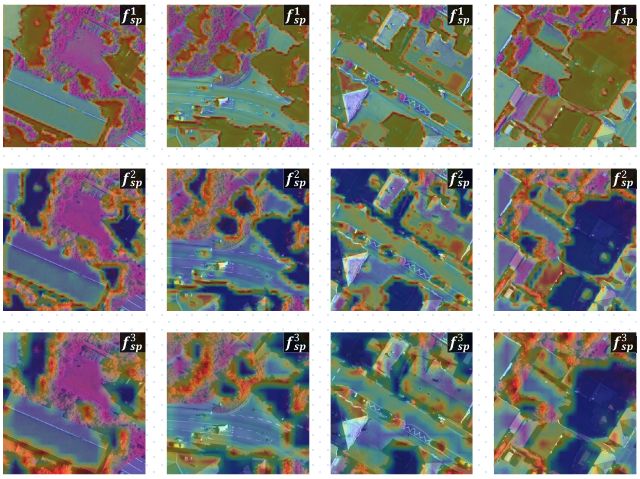


Fig. 6. Visualization of SPM features mentioned in (5) on cross-domain tasks P2V.

compact clusters in the feature space regardless of the input domain, indicating that Bridge effectively aligns the source and target domains. Although a clear domain gap exists between the source and target data, features from the same semantic category remain well clustered, demonstrating that Bridge successfully learns both domain-invariant and domain-specific representations. In addition, as highlighted by the orange circles, features from different semantic categories exhibit clear separation, forming distinct clustered regions, which further shows that Bridge possesses strong adaptation capability and can effectively capture category-specific boundaries.

6) Domain-Specific Tokens Visualization: To better understand what the domain-specific tokens have learned, we visualize their spatial correspondence to the RSIs. Since the tokens cannot be directly projected back to the pixel space, we use (10) to compute the feature–token similarity between each domain-specific token and the backbone feature map at multiple layers (layers 6, 12, 18, and 24). The similarity

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TABLE VI

COMPUTATIONAL COST AND PERFORMANCE COMPARISON. WE REPORT PARAMETERS, TRAINING GPU MEMORY USAGE (GB) ↓, TRAIN AND INFER SPEED (S/ITER) ↓, FLOPS ↓, FPS ↓, AND PERFORMANCE ↑ BETWEEN FULL FINE-TUNE, SIAMSEG, EARTH-ADAPTER, AND BRIDGE. ALL RESULTS ARE THE AVERAGE SCORE OF P2V, V2P AND R2U BENCHMARKS, DEMONSTRATING THE EXCELLENT TRADEOFF OF BRIDGE

Backbone	Method	Params	Memery(GB)	Train(s/iter)	Infer(s/iter)	Flops	FPS	mIoU(%)
Previous SOTA UDA methods								
MiT-B5 [17]	SiamSeg [22]	89.9M	10.29	0.69	**0.08**	**67.5**G	**12.5**	54.1
Cross-Domain PEFT Methods								
DINOv2 [28] (Large)	Full Fine-Tune	304.2M	21.54	0.80	0.10	310.6G	10.0	16.9
	+Earth-Adapter [55]	**7.1**M	15.94	0.71	0.12	398.4G	8.3	61.9
	+Bridge	8.3M	**11.96**	**0.58**	0.10	343.8G	10.0	**66.4**

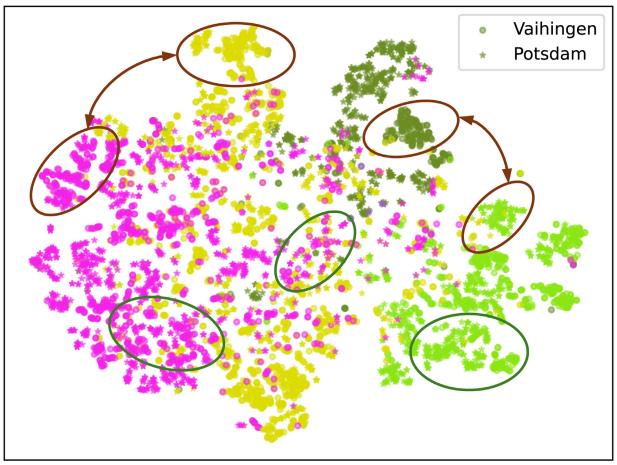


Fig. 7. T-SNE [60] embedding of the last layer of the backbone cross-domain tasks P2V. Dots and stars represent two different domains. Color-label mapping: dark green represents the impervious surface, light green represents the building, yellow represents the low vegetation, red represents the tree, and purple represents the car.

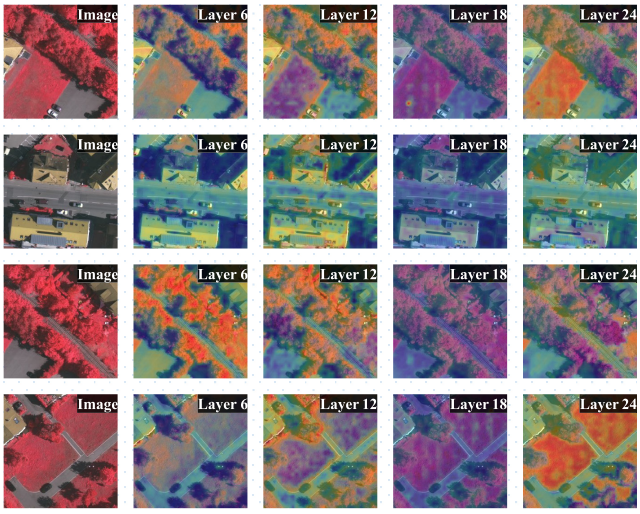


Fig. 8. Visualization of feature-token similarity mentioned in (10) on cross-domain tasks P2V.

between the token and each spatial feature vector is first computed, and then token-wise average pooling is applied across all tokens to obtain a spatial heatmap. Finally, the heatmaps are overlaid on the original images for interpretability. As shown in Fig. 8, tokens at different layers attend to different semantic structures: early tokens tend to respond to large-scale regions (e.g., vegetation and land parcels), while deeper tokens focus on more compact objects (e.g., roads and buildings). This indicates that domain-specific tokens encode multiscale spatial priors relevant to RS imagery.

7) Performance-Speed Tradeoff Ablation: As shown in Table VI, we compare the parameter counts, training GPU memory usage, training/inference speed, Flops, FPS, and performance of SiamSeg, Full Fine-Tune, Earth-Adapter, and the proposed Bridge. We only report training GPU memory usage, as training is the most memory-demanding stage for all methods. Inference memory is substantially smaller and highly correlated with training memory. Thus, reporting training memory sufficiently reflects practical resource overhead. The results indicate that, compared with the (DASS) method SiamSeg, which requires a large number of trainable parameters and incurs substantial training costs, both Earth-Adapter and Bridge achieve efficient and fast training with only a small fraction of parameters and lower training memory requirements while maintaining strong performance. Moreover, Bridge increases computational cost by only approximately 10% and reduces memory usage by nearly 10 GB relative to the DINOv2-L backbone, while delivering significantly stronger performance and effectively avoiding the performance collapse observed in Full Fine-Tune, which arises from the large domain gap between natural-image pretraining and downstream RS tasks. This highlights that Bridge achieves a significantly better accuracy–memory tradeoff.

8) Study on Different Data Preprocessing Methods: In our original experiments, Potsdam was preprocessed in the same manner as Earth-Adapter [55]. For Vaihingen, however, we adopted a stride of 256 instead of 512. The reason is that Vaihingen contains fewer and smaller images (average resolution 2906×2273), which yields significantly fewer training samples under a stride = 512. Reducing the stride to 256 increases sample diversity and improves optimization stability [22], which is a common practice in RS semantic segmentation.

To enable a fairer comparison, we additionally downloaded the processed dataset released by Earth-Adapter and retrained all experiments under the same training/validation splits and preprocessing settings as Earth-Adapter. The results are reported in Table VII. We observe that decreasing the number of training samples (due to a stride of = 512) slightly reduces the performance of all methods. Nevertheless, our proposed Bridge still achieves competitive or superior performance compared to Earth-Adapter. For example, under the Earth-Adapter dataset setting, Bridge improves over Earth-Adapter by +1.6 mIoU on the P2V task, confirming

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TABLE VII

STUDY ON DIFFERENT DATA PREPROCESSING METHODS. NOTE THAT THE CROPPING SIZE USED DURING DATA PREPROCESSING FOR VAI IS 512 × 512, WITH A STRIDE OF 512. ALL METHODS ARE BASED ON CROSS-DOMAIN TASK P2V

Backbone	Method	P2V	V2P	Avg.
DINOv2 [28]	+Adapter [46]	66.4	59.3	62.8
	+VPT [35]	66.2	59.3	62.7
	+Earth-Adapter [55]	67.7	62.2	64.9
DINOv2 [28]	+Bridge	**69.3**	**63.4**	**66.4**

TABLE VIII

STUDY ON γ SETTING. ALL METHODS ARE BASED ON CROSS-DOMAIN TASK P2V

	Backbone	Setting	mIoU(%)
1	DINOv2 [28]	$\gamma = 4.5 \times 10^{-1}$	73.4
2		$\gamma = 3.0 \times 10^{-1}$	73.8
3		$\gamma = 1.5 \times 10^{-1}$	**75.2**
4		$\gamma = 1.5 \times 10^{-2}$	74.0

TABLE IX

PERFORMANCE COMPARISON OF THE PROPOSED BRIDGE AGAINST DINOV2 SMALL, BASE, AND LARGE UNDER THE P2V DA SETTING. '+' MEANS THE BACKBONE IS FROZEN. THE BOLD FONT MEANS THE BEST PERFORMANCE

Backbone	Scale	Method	Params	mIoU(%)
DINOv2 [28]	Small	Full	22.1M	51.6
	Base	Full	86.5M	53.1
	Large	Full	304.2M	11.3
DINOv2 [28]	Large	+Bridge	**8.3M**	**75.2**

that our contributions remain valid and competitive under consistent data preprocessing.

9) Study on γ Setting: In our implementation, the parameter γ in (9) corresponds to the balancing coefficient, which regulates the tradeoff between maintaining domain-invariant knowledge from natural-image pretraining and injecting RS domain-specific spatial information. Intuitively, increasing γ increases the contribution of RS domain-specific cues while weakening the influence of domain-invariant features, whereas decreasing γ has the opposite effect. To analyze its impact, we perform a sensitivity experiment by fixing γ at different values and evaluating on the P2V setting, as shown in Table VIII. We observe that very small γ preserve more domain-invariant knowledge but fail to provide sufficient RS-specific cues, resulting in suboptimal domain alignment. Conversely, very large γ over-emphasizes domain-specific information and leads to the loss of useful pretrained knowledge, which also degrades performance. A moderate setting achieves the best balance, with the optimal result obtained at $\gamma = 1.5 \times 10^{-1}$, indicating that injecting a proper amount of RS domain-specific information is most beneficial for cross-DA.

VI. DISCUSSION

A. Full Fine-Tuning Collapse on Large VFMs

To identify the underlying cause of this phenomenon, we conducted an in-depth analysis across model scales, such as ViT-Small, ViT-Base, and ViT-Large, under identical hyperparameters and controlled optimization settings, including the same LR schedule, warmup, regularization, augmentations, and AdamW settings. As shown in Table IX, the results indicate that the collapse of ViT-L is not due to misconfigured hyperparameters but instead driven by model scale dependent optimization dynamics.

Under the same data condition (limited unlabeled target samples typical in RS UDA), ViT-L exhibits poor performance. In contrast, ViT-S and ViT-B exhibit competitive performance. These observations align with model scaling laws: larger models require disproportionately more data or iterations to reach a stable optimum, especially when the downstream dataset is small.

RSIs differ substantially from NIs in spectral patterns, geometry, and object scales. Larger models possess stronger inductive biases that are tightly coupled with the pretraining domain. When fine-tuned without sufficient target supervision, ViT-L tends to overfit its pretrained source priors, failing to adapt to RS-specific structures. This leads to optimization stagnation and eventual collapse. We emphasize that full fine-tuning works well for ViT-S and ViT-B. Only when scaling up to ViT-L do we observe a reproducible collapse, indicating that the issue emerges from the interaction between model size and severe domain discrepancy. This is precisely the limitation our method (Bridge) aims to address, that is, stabilizing large VFM adaptation without relying on extensive labeled data or costly full fine-tuning and eliminating domain discrepancy.

TABLE X

PERFORMANCE COMPARISON OF THE PROPOSED SPM UNDER THE P2V DA SETTING. † INDICATES THE SIMPLE FUSION OF SPM FEATURES WITH INTERMEDIATE FEATURES FROM VFM. ⋆ INDICATES THE USE OF DOMAIN-INVARIANT TOKEN T_i INSTEAD OF DOMAIN-SPECIFIC TOKENS $\tilde{T}_i$ FOR FEATURE REFINEMENT WITH VFM INTERMEDIATE FEATURES

	Backbone	SPM	DITA	DAI	mIoU(%)
1†	DINOv2 [28]	used	-	-	50.3
2⋆		-	-	used	60.2
3	DINOv2 [28]	used	used	used	**75.2**

B. Discussion on SPM

There is a common concern: does the gain come from the sophisticated DITA/DAI mechanism, or simply from the inclusion of the SPM? Indeed, SPM provides spatial priors that are beneficial for RS tasks. But the SPM (a lightweight CNN) extracts structural cues such as edges, textures, and local geometry. While useful, these priors do not address the fundamental domain discrepancy between NI-pretrained VFMs and RSI. In that case, SPM only provides low-level spatial bias, not domain alignment. As shown in Table X, when we replace DITA/DAI with a simple fusion of SPM features with the VFM feature, the model shows only marginal improvement and still suffers from significant degradation on the target domain. This indicates that CNN priors alone are insufficient to perform cross-DA.

In contrast, DITA converts the globally trained domain-invariant tokens into domain-specific ones by conditioning them on spatial priors. DAI then injects these adapted tokens into each VFM layer, refining the representation while keeping

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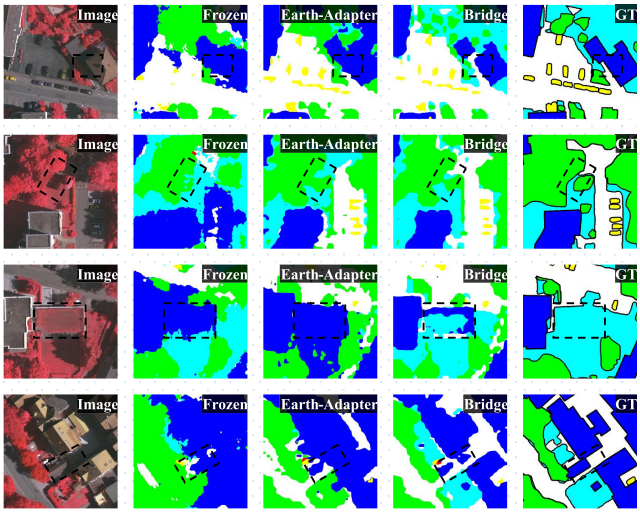


Fig. 9. Failure modes of Bridge on cross-domain task P2V.

the backbone frozen. Without these two mechanisms, the model fails to realign pretrained knowledge to the RS domain.

C. Discussion on Domain-Invariant Tokens

These tokens are designed to encode transferable priors that remain stable across natural and RS domains, functioning as a "bridge" that allows the VFM to gradually move from NI statistics toward RSI representations. Unlike prompts [35], [61] or adapters [46] that inject task information at fixed locations, our domain-invariant tokens interact dynamically with multiscale spatial priors (via DITA) and thus learn a hybrid representation that captures both domain-shared and domain-specific cues. Subsequently, the intermediate features of VFM are refined (via DAI) using the obtained domain-specific tokens.

D. Discussion on DITA and DAI

DITA computes token → prior similarity because the domain-invariant tokens serve as query-like carriers that selectively absorb spatial structural cues before adaptation. DAI, in contrast, computes feature → token similarity to enable progressive refinement that remains aligned with the hierarchical representations within the VFM.

Both DITA and DAI draw inspiration from the similarity computation mechanism in [62]. However, unlike the heavy multihead attention employed in [62], the similarity operations used in Bridge are deliberately lightweight. This design substantially reduces computational overhead and memory consumption while preserving the ability to effectively propagate domain-relevant information through the VFM layers.

E. Compare With Rein

While Rein [51] also includes a feature refinement module, it merely utilizes learnable tokens to refine features. Due to the substantial domain gap between NIs and RSIs, such a simple token-based refinement is insufficient to effectively transfer the pretrained knowledge of natural-image VFMs to the RS domain, which limits its performance. In contrast, our proposed DAI leverages multiscale spatial priors extracted by the SPM to guide the adaptation of domain-invariant tokens into domain-specific tokens tailored for the RS domain. These domain-specific tokens are then used to refine intermediate features, providing richer prior knowledge of the target domain. This design is empirically validated in Table X. When only simple token refinement is applied (Method 2, using domain-invariant tokens without DITA), the model achieves 60.2% mIoU. By incorporating both SPM and DITA to produce domain-specific tokens for DAI (Method 3), the mIoU increases to 75.2%, demonstrating that DAI substantially improves feature adaptation by leveraging RS priors, compared to the simple token refinement approach in Rein.

F. Limitations and Future Works

1) Limitations: Although DASS methods can achieve significant performance improvements on cross-domain tasks, they may still exhibit recognition inaccuracies in certain scenarios. As shown in Fig. 9, we visualize several representative failure cases to better reveal the limitations of the proposed framework. We observe that most failures occur in visually ambiguous regions with low interclass contrast (e.g., shadows or occlusions), where even well-aligned domain-invariant features remain insufficient to fully resolve category confusion. These failure modes indicate that, despite substantial progress in cross-domain alignment, bridging the semantic gap between natural-image pretraining and RS imagery remains inherently challenging in certain scenes. In such cases, collecting a small amount of target-domain annotations becomes crucial, as it can further enhance the model's ability to distinguish ambiguous regions and reduce semantic confusion. Moreover, although DASS effectively mitigates performance degradation caused by domain shift, it requires joint training over both source and target data. This implies that for different adaptation scenarios, the model must be retrained, which introduces additional computational overhead. When the source-domain scale becomes large, retraining for every new target domain may quickly become impractical.

2) Future Works: A promising direction to alleviate the above limitations is to investigate domain generalization semantic segmentation (DGSS) techniques for RS semantic segmentation. Unlike DASS, which requires joint access to source and target data, DGSS aims to learn models that can generalize to unseen domains without target-domain exposure during training. This paradigm not only reduces the retraining cost in multidomain scenarios but also better reflects real-world deployment conditions, where RS imagery exhibits diverse geographical, seasonal, and sensor-induced variations. Following existing DGSS studies [55], incorporating Bridge into a DGSS training paradigm is a promising future direction. We plan to explore the integration of Bridge into DGSS frameworks to improve semantic segmentation robustness under unseen domain conditions.

VII. Conclusion

In this article, we first highlight the performance degradation observed when applying existing PEFT methods to adapt NIs pretrained VFMs to RS downstream tasks, such as DASS. We

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attribute this issue to the substantial intradomain gap between NI and RSIs, as well as the inner-domain gap inherent to RS DASS tasks. These factors hinder existing PEFT approaches from effectively transferring pretrained knowledge to the inner-target domain. While pretraining models from scratch in the RS domain could mitigate this issue, it incurs prohibitive computational and annotation costs. To address this challenge, we propose Bridge, a PEFT method tailored for RS DASS. The Bridge is designed to effectively transfer knowledge from NI-pretrained VFMs while adapting to the unique characteristics of RSI. In our experiments, we first compare Bridge with state-of-the-art DASS methods and mainstream PEFT approaches, demonstrating its superiority. We then conduct a detailed analysis of the contributions of individual components within Bridge, as well as its effectiveness when combined with different VFMs. Furthermore, we provide qualitative visualizations and examine training and inference efficiency. Overall, the experimental results verify that Bridge achieves superior performance while maintaining robustness, compatibility, and efficiency, thereby offering a practical solution for adapting VFMs to RS DASS tasks.

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